

Section I Multiple Choice (10 marks)

Attempt Question 1 – 10 (1 mark each)

Allow approximately 15 minutes for this section.

Question 1

An infinite geometric series has a first term of 8 and a limiting sum of 12. What is the common ratio?

- A) $1/6$ B) $5/3$ C) $1/2$ D) $1/3$

Question 2

What is the greatest value taken by the function $f(x) = 4 - 2\cos x$ for $x \geq 0$?

- A) 2 B) 4 C) 6 D) 8

Question 3

What is the value of $\int_2^6 \frac{1}{x+2} dx$?

- A) $\ln 2$ B) $\ln 4$ C) $\ln 6$ D) $\ln 8$

Question 4

The table below shows the values of a function $f(x)$ for five values of x .

x	2	2.5	3	3.5	4
$f(x)$	4	1	-2	3	8

What value is an estimate for $\int_2^4 f(x)dx$ using Simpson's Rule with these five values ?

- A) 4 B) 6 C) 8 D) 12

Question 5

It is known that the number $N(t)$ of ants in a certain nest at time $t \geq 0$ is given by $N(t) = \frac{K}{1+e^t}$ where K is constant and t is measured in months.

At time $t = 0$, $N(t)$ is estimated at 2×10^5 ants. What is the value of K ?

- A) 2×10^5 B) 2×10^{-5} C) 4×10^5 D) 4×10^{-5}

Question 6

Sixty tickets are sold in a raffle. There are two prizes. Lincoln buys 5 tickets. Which expression gives the probability that Lincoln wins both prizes ?

- A) $\frac{5}{60} + \frac{4}{59}$ B) $\frac{5}{60} + \frac{4}{60}$ C) $\frac{5}{60} \times \frac{4}{59}$ D) $\frac{5}{60} \times \frac{4}{60}$

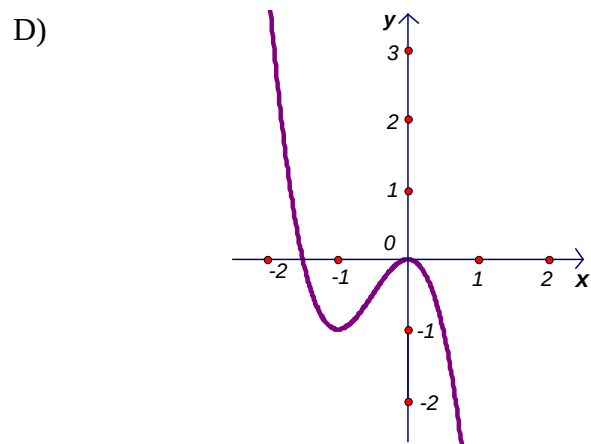
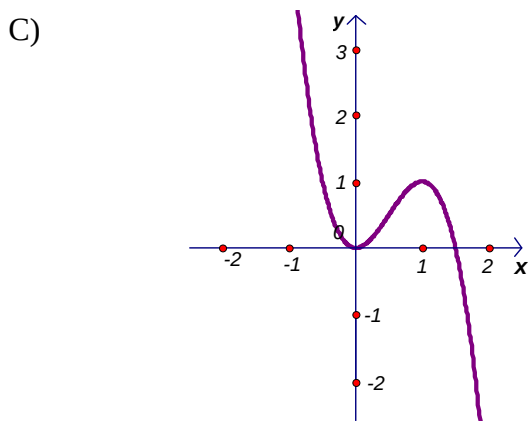
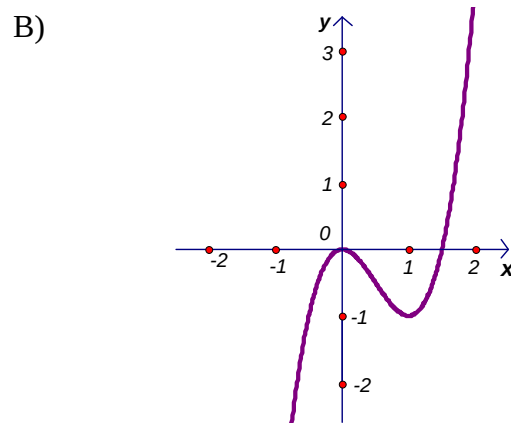
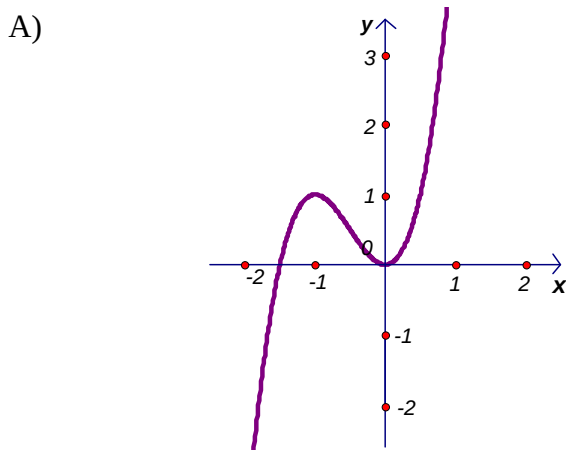
Question 7

What is the equation of the **normal** to the curve $y = x^2 - 4x$ at $(1, -3)$?

- A) $x + 2y - 7 = 0$ B) $x - 2y - 7 = 0$
C) $2x - y - 5 = 0$ D) $2x + y + 5 = 0$

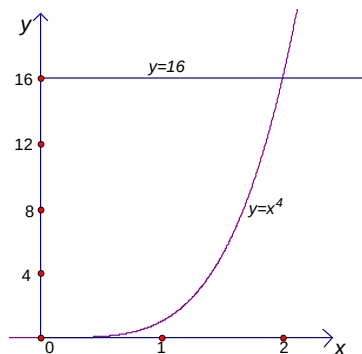
Question 8

Which of the following is the graph of $y = 2x^3 - 3x^2$?



Question 9

The region in the diagram is bounded by the curve $y = x^4$, the y-axis and the line $y = 16$.



Which of the following expressions is correct for the volume of the solid of revolution when this region is rotated about the y-axis?

- A) $\pi \int_0^2 x^8 dx$ B) $\pi \int_0^{16} x^8 dx$
C) $\pi \int_0^2 \sqrt{y} dy$ D) $\pi \int_0^{16} \sqrt{y} dy$

Question 10

What are the solutions to the equation $e^{6x} - 7e^{3x} + 6 = 0$?

- A) $x = 1$ or $x = 6$ B) $x = 0$ or $x = \frac{\ln 6}{2}$
C) $x = 0$ or $x = \frac{\ln 6}{3}$ D) $x = 1$ or $x = \frac{\ln 6}{2}$

End of Section I

Section II Total Marks is 90

Attempt Questions 11 – 16.

Allow approximately 2 hours & 45 minutes for this section.

Answer all questions, starting each new question on a new sheet of paper with your **student ID number** in the top right hand corner and the question number on the left hand side of your paper.

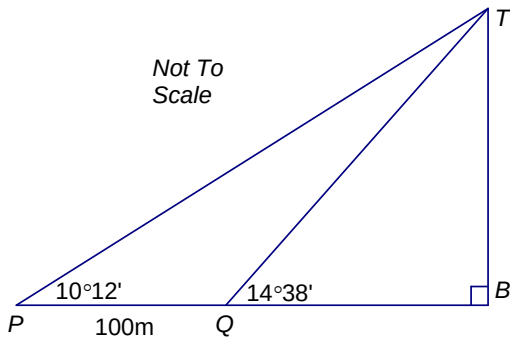
All necessary working must be shown in each and every question.

Question 11 begins on the next page

Question 11 (15 Marks)

Start a new piece of paper

Marks

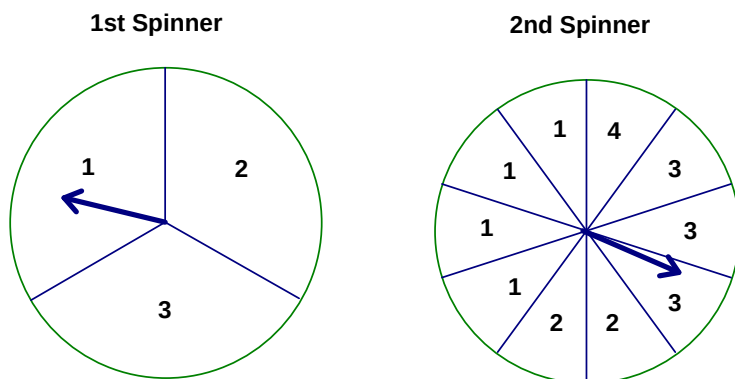
- a) Simplify $\frac{3x}{x+2} - \frac{5x+19}{x^2+5x+6}$ 2
- b) Differentiate i) $y = \ln\left(\frac{x-1}{x^2}\right)$ ii) $y = x^2 \cos 4x$ 4
- c) Integrate i) $\int 3x + e^{4x} dx$ ii) $\int \frac{x^2-1}{x} dx$ 4
- d) The angle of elevation of the top of tree BT when viewed from point P is $10^\circ 12'$.
After walking 100m directly towards the tree one arrives at Q where the angle of elevation is $14^\circ 38'$.
Copy the diagram and find the height of the tree to the nearest centimetre. 3
- 
- e) Make a careful sketch of the curve $y = 1 + 3\sin \frac{x}{2}$ over the domain $0 \leq x \leq 2\pi$. 2

Question 12 (15 Marks)

Start a new piece of paper

Marks

- a) The diagram below shows two spinners.



Each of the three outcomes on the first spinner are equally likely. On the second spinner there are ten equally likely sectors for the arrow to land on with four possible outcomes.

In a game, both spinners are spun simultaneously. The player's score is the **sum** of the two numbers that the spinners land on. (eg A score of 4 in the above diagram). A player wins if their score is an odd number greater than 4.

- i) What is the probability of scoring 7? 1
- ii) What is the probability that a player will win the first game? 3
- iii) What is the probability that a player will win the first three games? 1

Question 12 is continued on the next page

Question 12 (continued)

- b) The fourth term of an Arithmetic Sequence is (-12) and the tenth term is 21.
Calculate the value of the nineteenth term. 2
- c) The point $Q(-2,1)$ lies on the line k which has equation $9x - 2y + 20 = 0$.
The point $R(4,-2)$ lies on the line l which has equation $3x + y - 10 = 0$.
- i) Find the coordinates of P , the point on the y -axis where k and l intersect. 2
- ii) The line m joins Q and R . Show that the equation of m is $x + 2y = 0$. 2
- iii) Show, by shading on a sketch, the region defined by the three inequalities:
 $9x - 2y + 20 \geq 0$, $3x + y - 10 \leq 0$, $x + 2y \geq 0$. 1
- iv) Find, as a surd, the perpendicular distance from P to m and hence, or otherwise, find the exact area of the triangle bounded by the lines k , l and m . 3

Question 13 (15 Marks)**Start a new piece of paper****Marks**

- a) i) By completion of the square, or otherwise, show that the vertex of the parabola $x^2 - 10x + 15 = 2y$ is at $(5, -5)$ 1
- ii) Write down the focus of this parabola. 2
- iii) Show that the parabola $y = 4x - x^2$ also passes through the point $(5, -5)$. 1
- iv) Find the other point of intersection of these two parabolas. 2
- v) Hence find the area enclosed between the two parabolas. 3
- b) The velocity, $\dot{x}$, in m/s of a particle moving in a straight line is given by
 $\dot{x} = 1 - 2 \sin t$ for $0 \leq t \leq 2\pi$, where t is the time in seconds. The particle is initially at $x = 2$.
- i) At what time(s) is the acceleration zero? 1
- ii) What is the maximum velocity of the particle during this period.
(You should demonstrate that this is a maximum and not a minimum.) 2
- iii) Find the first time that the particle changes direction during this period. 1
- iv) Hence, or otherwise, find the exact distance travelled by the particle between $t = 0$ and the time when the particle first changes direction. 2

Question 14 (15 Marks)

Start a new piece of paper

Marks

- a) A sheep is grazing in a large paddock which is bounded on one side by a long straight fence. The sheep is tethered to a stake by a rope 20m in length. If the stake is placed 10m from the fence, find the area to the nearest square metre over which the sheep can graze.

3

- b) The quadratic equation $x^2 - 3x - 13$ has roots α and β .

i) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

1

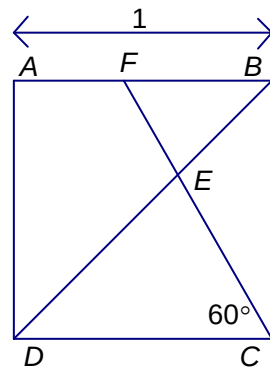
ii) What is the exact value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$?

2

iii) What is the value of $6\beta - 2\beta^2$?

1

- c) A square $ABCD$ of side length 1 unit is shown below. The point F is drawn on AB such that $\angle DCF = 60^\circ$. The diagonal DB intersects CF at E .



i) Show that $\triangle DEC \parallel \triangle BEF$

3

ii) Show that $FB = \frac{1}{\sqrt{3}}$

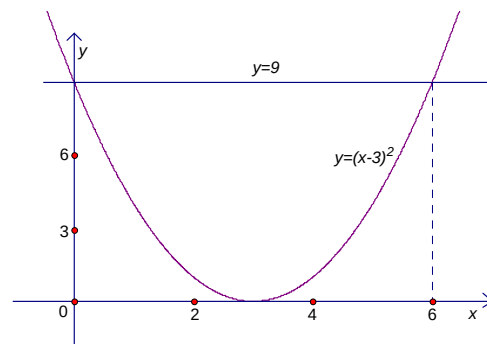
1

iii) Hence, or otherwise, find the ratio $\text{Area}\triangle DEC : \text{Area}\triangle BEF$

1

- d) The region bounded by the graph $y = (x - 3)^2$ and the line $y = 9$ is rotated about the x-axis to form a solid of revolution.

Find the volume of the solid so formed.



3

Question 15 (15 Marks)**Start a new piece of paper****Marks**

- a) Following an accident, water started leaking out of a tank. If the volume of water in the tank was $V(t)$ litres, then t days after the accident, the rate of change of V was given by $\frac{dV}{dt} = 20t - 300$ litres per day.

When the tank stopped leaking, it still had 4750 litres in it.

- i) For how many days was the tank leaking? **1**
- ii) Find a formula for V in terms of t . **3**
- iii) How much water was in the tank when it started leaking? **1**

- b) Allcare Home Loans has a special package for first home buyers. The main details of the package, as shown in their brochure, are summarised in the table.

Stage	Term	Special Features	Interest Rate
Introductory Stage	0-2 years (2 years)	No monthly repayments	6% pa compounded monthly
Secondary Stage	2-10 years (8 years)	Monthly repayments start. At the end of this period, the amount owing must be reduced to the original size of the loan.	9% pa compounded monthly
Final Stage	Variable (but not exceeding 20 years)	The borrower determines the size of the monthly repayment, provided that the loan is repaid within 20 years from the start of this stage.	12% pa compounded monthly

Alice and Bernard have accepted the terms of the above plan and they have borrowed \$500,000 to finance their first house.

- i) Show that the amount owing at the end of the Introductory Stage, to the nearest dollar, is \$563,580. **1**
- ii) The principal for the Secondary Stage will be \$563,580. Assume that the first monthly repayment, M , is paid after one month into the Secondary period. Find M , to the nearest cent if the amount owing at the end of the Secondary Stage is to be \$500,000. **4**
- iii) At the start of the Final Stage, Alice and Bernard have decided that they can afford to repay \$6500 per month.
 - α) Determine how many full payments of \$6500 it will take for the loan to have been repaid in full. **3**
 - β) The last monthly repayment of \$6500 is more than required. How much should be refunded to Alice and Bernard? **2**

Question 16 (15 Marks)**Start a new piece of paper****Marks**

- a) Given the function $y = \frac{10}{3+2\cos x}$ in the domain $0 \leq x \leq 2\pi$
- i) Find the location and nature of all the stationary points in the domain. **3**
 - ii) Graph the function in the given domain. **2**
- b) A mining company simultaneously established three new mining towns, A , B and C . Each had an initial population of 500 and it was planned that they each would grow by 50 inhabitants per year for the first ten years.
- i) Only town A grew as planned. Write down an expression for the intended population of town A , t years after its opening ($t \leq 10$ and t is an integer). **2**
 - ii) For various reasons, towns B and C did not grow as planned. Their populations are better modelled by:

Town B : $\frac{dP_B}{dt} = -0.3P_B$

Town C : $P_C = 100\left(5 + t - \frac{t^2}{4}\right)$
 - α) Show that the expression $P_B = 500e^{-0.3t}$ satisfies the equation describing town B . **1**
 - β) Calculate, to the nearest integer, the population of town B after 6 years. **1**
 - γ) Find when the population of town C reached its maximum and what was that maximum value? **3**
 - δ) The mining company has determined that any town is unviable if the population goes below 50. Which will be the first town to close? (Justify your answer) **3**

END OF EXAM

JRAHS 20 MATHS TRIAL, 2012

MATHEMATICS: Question.....		MULTIPLE CHOICE
Suggested Solutions	Marks	Marker's Comments
520 = 12 1. $\frac{8}{1-1} = 12$ $\therefore r = \frac{1}{3}$ (D) 2. $-1 \leq \cos x \leq 1$ $\Rightarrow 4 - 2\cos x \leq 6$ (C) 3. $\left[\ln(x+2) \right]_2^6 = \ln \frac{8}{4} = \ln 2$ (A) 4. $A = \frac{0.5}{3} \{ 4 + 4 \times (1) + (-2) \} +$ $\frac{0.5}{3} \{ -2 + 4 \times 3 + 8 \}$ $= 4$ (A) 5. $2 \times 10^5 = \frac{k}{1+e^0}$ $2 \times 10^5 = \frac{k}{2} \therefore k = 4 \times 10^5$ (C) 6. $\frac{5}{60} \times \frac{4}{50}$ (C) 7. $y = x^2 - 4x$ $y' = 2x - 4$ at $x = 1$ $m_t = -2$ $\therefore m_n = \frac{1}{2}$ (B) $y+3 = \frac{1}{2}(x-1)$ $x-2y-7=0$ 8. $y = 2x^3 - 3x^2$ $= x^2(2x-3)$ $\Rightarrow$ double root at $x=0$ & a single root at $x = \frac{3}{2}$ when $x=1$ $y = -1 \Rightarrow$ (B) 9. $V = \int_0^{16} \pi x^2 dy$ $= \pi \int_0^{16} \sqrt{y} dy$ (D) 10. $e^{6x} - 7e^{3x} + 6 = 0$ let $u = e^{3x}$ $u^2 - 7u + 6 = 0$ $(u-6)(u-1) = 0$ $e^{3x} = 6$ or $e^{3x} = 1$ $3x = \ln 6$ $3x = \ln 1$ $x = \frac{\ln 6}{3}$ $x = 0$ (C)		

MATHEMATICS: Question 11

1

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} (a) \quad & \frac{3x}{x+2} - \frac{5x+19}{(x+2)(x+3)} \\ &= \frac{3x(x+3) - 5x - 19}{(x+2)(x+3)} \\ &= \frac{3x^2 + 4x - 19}{(x+2)(x+3)} \end{aligned}$$

1

($\frac{1}{2}$ off for each error)

Students still have difficulty expanding with negative signs.

[2]

$$\begin{aligned} (b)(i) \quad y &= \ln \frac{(x-1)}{x^2} \\ &= \ln(x-1) - \ln x^2 \\ &= \ln(x-1) - 2 \ln x \end{aligned}$$

($\frac{1}{2}$ off for each error)

$$\begin{aligned} y' &= \frac{1}{x-1} - \frac{2}{x} \\ &= \frac{2-x}{x(x-1)} \end{aligned}$$

2

Students used quotient rule instead of using log laws.

[2]

$$\begin{aligned} (ii) \quad y &= x^2 \cos 4x \\ \frac{dy}{dx} &= 2x \cdot \cos 4x + x^2 \cdot (-4 \sin 4x) \\ &= 2x(\cos 4x - 2x \sin 4x) \end{aligned}$$

2

[2]

$$\begin{aligned} (c)(i) \quad & \int 3x + e^{4x} dx \\ &= \frac{3x^2}{2} + \frac{e^{4x}}{4} + C \end{aligned}$$

2

($-\frac{1}{2}$ if no constant)

[2]

$$\begin{aligned} (ii) \quad & \int \frac{x^2-1}{x} dx \\ &= \int \frac{x^2}{x} - \frac{1}{x} dx \\ &= \int x - \frac{1}{x} dx \\ &= \frac{x^2}{2} - \ln x + C \end{aligned}$$

2

($-\frac{1}{2}$ for each error)

[2]

Students failing to make good use of S.I. sheet.

Suggested Solutions

Marks

Marker's Comments

Method 1

- (1) $\angle PQT + 14^\circ 38' = 180^\circ$
 (2) (sum of straight angle $PQB = 180^\circ$)
 $\therefore \angle PQT = 165^\circ 22'$
 (1) $\angle PTQ + 10^\circ 12' + 165^\circ 22' = 180^\circ$
 (2) (angle sum of $\triangle PTQ = 180^\circ$)

(1) $\frac{PT}{\sin 165^\circ 22'} = \frac{100}{\sin 4^\circ 26'}$

$\sin 10^\circ 12' = \frac{BT}{PT}$

(1) $\therefore BT = PT \sin 10^\circ 12'$
 $= \frac{100 \sin 165^\circ 22' \cdot \sin 10^\circ 12'}{\sin 4^\circ 26'}$
 $= 57.8756...$

(1) Thus, the height of the tree is
 (2) 57.88 m (to nearest cm)

Method 2

- (1) $\angle PTB + 90^\circ + 10^\circ 12' = 180^\circ$
 (2) (angle sum of $\triangle PTB = 180^\circ$)
 $\therefore \angle PTB = 79^\circ 48'$
 Let $h = BT$ and $x = QB$
 (1) $\tan 79^\circ 48' = \frac{100 + x}{h}$ — (1)

- (1) $\angle QTB + 14^\circ 38' + 90^\circ = 180^\circ$
 (2) (angle sum of $\triangle QTB$ is 180°)
 $\therefore \angle QTB = 75^\circ 22'$

$\tan 75^\circ 22' = \frac{x}{h}$ — (2)

(1) From (2) $x = h \tan 75^\circ 22'$

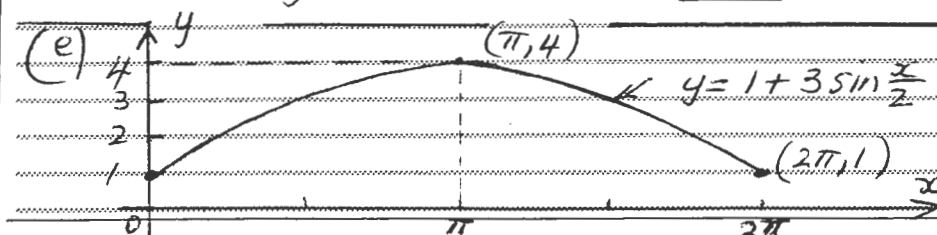
subst. into (1)

$\tan 79^\circ 48' = \frac{100 + h \tan 75^\circ 22'}{h}$

$h (\tan 79^\circ 48' - \tan 75^\circ 22') = 100$

$h = \frac{100}{\tan 79^\circ 48' - \tan 75^\circ 22'}$

(1) $= 57.8756...$
 $\therefore$ Height of the tree is 57.88 (nearest cm).



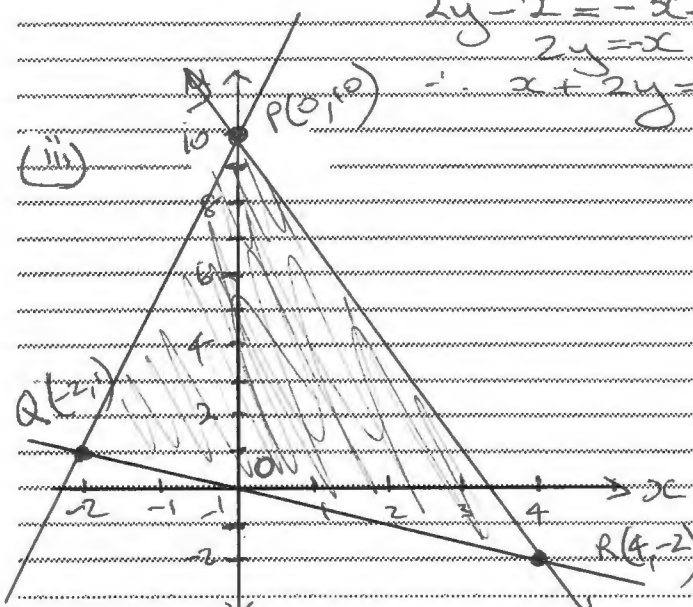
$\frac{1}{2}$ each for the max. pt & endpts.
 $\frac{1}{2}$ equation
 $-\frac{1}{2}$ incorrect curvature.

MATHEMATICS: Question 1.2.

Suggested Solutions	Marks	Marker's Comments
(a)(i) $P(\text{score of } 7) = P(3 \text{ and } 4)$ $= \frac{1}{3} \times \frac{1}{10}$ $= \frac{1}{30}$	1	
(ii) $P(\text{win 1st game}) = P(5) \text{ or } P(7)$ $= P(1+4) + P(2+3) + P(3+2) + P(4+1)$ $= \left(\frac{1}{3} \times \frac{1}{10}\right) + \left(\frac{1}{3} \times \frac{2}{10}\right) + \left(\frac{1}{3} \times \frac{2}{10}\right) + \frac{1}{30}$ $= \frac{7}{30}$	1 mark 1 mks 1 mark	* If they got an answer of $\frac{1}{5}$ $\therefore$ 2 mks max. * If they got $\frac{1}{5}$ but they had the first line $\therefore$ 2 1/2 mks.
(iii) $P(\text{win 1st 3 games}) = \frac{7}{30} \times \frac{7}{30} \times \frac{7}{30}$ $= \frac{343}{27000}$	1	
(b) $a + 3d = -12$ (1) $a + 9d = 21$ (2) $(2) - (1)$ $6d = 33$ $d = \frac{11}{2} = 5\frac{1}{2}$ sub into (1) $a + 3(5\frac{1}{2}) = -12$ $a = -28\frac{1}{2}$ $T_n = a + (n-1)d$ $= -28\frac{1}{2} + 18 \times 5\frac{1}{2}$ $= 70\frac{1}{2}$	1/2 1/2 1/2 1/2	* A lot of students did NOT know the formula $\therefore$ * If they left off the "-" sign for (1) lost 1mk. * wrong formula = 0 mks
(c)(i) $9x - 2y + 20 = 0$ (1) $3x + y - 10 = 0$ $6x + 2y - 20 = 0$ (2) $(1) + (2)$ $15x = 0$ $x = 0$ sub into (1) $0 - 2y + 20 = 0$ $2y = 20$ $y = 10$ $\therefore P \text{ is } (0, 10)$	2 mks	

2/2

MATHEMATICS: Question 12

Suggested Solutions	Marks	Marker's Comments
(i) $m_{QR} = \frac{2-1}{2-2} = \frac{1-2}{2-4} = \frac{3}{-6} = -\frac{1}{2}$ eqn of QR: $y - y_1 = m(x - x_1)$ $y - 1 = -\frac{1}{2}(x + 2)$ $2y - 2 = -x - 2$ $2y = -x$ $\therefore x + 2y = 0$ (ii)  (iii) $d = \frac{ ax + by + c }{\sqrt{a^2 + b^2}}$ $= \frac{ 0 \times 1 + 2 \times 10 + 0 }{\sqrt{0^2 + 2^2}}$ $= \frac{20}{\sqrt{5}}$ $= 4\sqrt{5} \text{ units}$ QR = $\sqrt{(4 - (-2))^2 + (-2 - 1)^2}$ $= \sqrt{36 + 9}$ $= \sqrt{45}$ $= 3\sqrt{5}$ Area = $\frac{1}{2}bh$ $= \frac{1}{2} \times 3\sqrt{5} \times 4\sqrt{5}$ $= 30 \text{ units}^2$	1 1 1 1 1 1	It was a proof so you had to show working to justify the marks!! *All students needed to do was plot P, Q and R but..... *students forget how to simplify surds!!! $\sqrt{9} = 3$

20 MATHS TRIAL SOLUTIONS, 2012

MATHEMATICS: Question 13

Suggested Solutions

Marks

Marker's Comments

(a)(i) $x^2 - 10x + 15 = 2y$

METHOD I: $x^2 - 10x + 25 = 2y - 15 + 25$

$(x-5)^2 = 2y + 10$

$(x-5)^2 = 2(y+5)$

As it is of the form $(x-h)^2 = 4a(y-k)$
 $\therefore$ Vertex $V = (h, k) = (5, -5)$ ed.

METHOD II: $y = \frac{1}{2}x^2 - 5x + \frac{15}{2}$

Axis of symmetry $x = -\frac{b}{2a} = -\frac{-5}{2 \times \frac{1}{2}} = 5$

at $x = 5$ $25 - 10 \times 5 + 15 = 2y$

$2y = -10$

$y = -5$

$\therefore$ T. point $= V(5, -5)$

(ii) As $4a = 2$

$\therefore a = \frac{1}{2}$

$\therefore$ focus $S = (5, -5 + \frac{1}{2}) = (5, -4\frac{1}{2})$

(iii) Verify $(5, -5)$ satisfies $y = 4x - x^2$

LHS

$y = -5$

RHS

$4x - x^2 = 4 \times 5 - 5^2$

$= 20 - 25$

$= -5$

$\therefore$ LHS = RHS

$\therefore (5, -5)$ also lies on $y = 4x - x^2$

(iv) $y = 4x - x^2$ (1)

$2y = x^2 - 10x + 15$ (2)

$\therefore 2(4x - x^2) = x^2 - 10x + 15$

$8x - 2x^2 = x^2 - 10x + 15$

$0 = 3x^2 - 18x + 15$

$(\div 3)$ $x^2 - 6x + 5 = 0$

$(x-1)(x-5) = 0$

$\therefore x = 1$ or 5

in eqn (1) $y = 3$ or -5

$\therefore$ Other point is $(1, 3)$

III

2

II

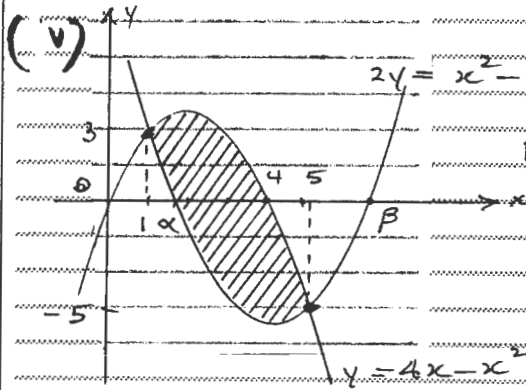
2

Suggested Solutions

Marks

Marker's Comments

(v)



$2y = x^2 - 10x + 15$
 $y = 4x - x^2 = y_1$
 $y = x^2 - 10x + 15 = y_2$

$$\text{AREA} = \int_1^5 y_2 - y_1 \, dx$$

$$= \int_1^5 (x^2 - 10x + 15) - (4x - x^2) \, dx$$

$$= \int_1^5 (2x^2 - 14x + 15) \, dx$$

$$= \left[\frac{2}{3}x^3 - 7x^2 + 15x \right]_1^5$$

$$= \left(\frac{2}{3}(125) - 7(25) + 15(5) \right) - \left(\frac{2}{3}(1) - 7(1) + 15(1) \right)$$

$$= \left(\frac{250}{3} - 175 + 75 \right) - \left(\frac{2}{3} - 7 + 15 \right)$$

$$= \left(\frac{250}{3} - 100 \right) - \left(\frac{2}{3} + 8 \right)$$

$$= \frac{250}{3} - 100 - \frac{2}{3} - 8$$

$$= \frac{250 - 300 - 2 - 24}{3}$$

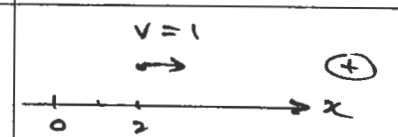
$$= \frac{-76}{3}$$

$\therefore \text{Area} = \frac{76}{3}$ sq units

(b) (i) $x = y = 1 - 2 \sin t$ $0 \leq t \leq 2\pi$

Date $t = 0$ $x = 2$ $v = 1$ $\ddot{x} = -2$

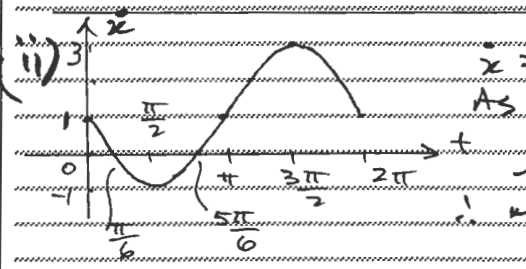
$\ddot{x} = -2 \cos t = 0$
 $\therefore \cos t = 0$
 $t = \frac{\pi}{2}$ or $\frac{3\pi}{2}$
 $\therefore$ accel. is zero at $\frac{\pi}{2}$ or $\frac{3\pi}{2}$ seconds.



(ii) $x = 1 - 2 \sin t$

As $-1 \leq \sin t \leq 1$
 $\therefore -1 \leq 1 - 2 \sin t \leq 3$
 $\therefore$ max speed is 3 m/s at $t = \frac{3\pi}{2}$

OR use Calculus using (b) (i) $t = \frac{\pi}{2}, \frac{3\pi}{2}$ and TEST



(iii) Particle at REST only $v = \dot{x} = 0$

$1 - 2 \sin t = 0$
 $\sin t = \frac{1}{2}$
 $t = \frac{\pi}{6}$ or $\frac{5\pi}{6}$
 First $t = \frac{\pi}{6}$ (seconds)

TEST if changes direction $\ddot{x}(\frac{\pi}{6}) = -\sqrt{3} < 0$ $\therefore$ pushed back to left
 see sketch table: $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \pi$

$\frac{1}{2}$ for $\frac{\pi}{6}$
 $\frac{1}{2}$ for TEST.

(iv) distance travelled $= \int_0^{\pi/6} (1 - 2 \sin t) \, dt$

$= [t + 2 \cos t]_0^{\pi/6}$
 $= \left(\frac{\pi}{6} + 2 \cos \frac{\pi}{6} \right) - (0 + 2)$
 $= \left(\frac{\pi}{6} + \sqrt{3} - 2 \right) \text{ m}$

DOES NOT CHANGE direction in $0 \leq t \leq \frac{\pi}{6}$

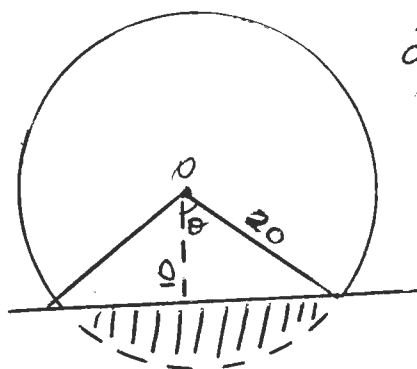
SEE (ii)

MATHEMATICS: Question...14.

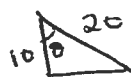
Suggested Solutions

Marks

Marker's Comments



a) step1 Get θ .



$$\cos \theta = \frac{10}{20}$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

step2 Find 2θ .

$$\therefore 2\theta = \frac{2\pi}{3} \checkmark$$

$$\text{shaded Sector} = \frac{1}{2} r^2 (\theta - \sin \theta)$$

$$= \frac{20^2}{2} \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3} \right)$$

$$= 200 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \checkmark$$

$$\therefore \text{Grazing area} = A_{\text{circle}} - A_{\text{segment}}$$

$$= \pi \times 20^2 - \left(200 \left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2} \right) \right)$$

$$= 400\pi - \frac{400\pi}{3} + 100\sqrt{3}$$

$$= 1010.96 \dots \checkmark$$

$$\text{Area is } 1011 \text{ m}^2 \text{ (to nearest m}^2\text{)}$$

$$\text{accepted } \frac{800\pi}{3} + 100\sqrt{3}.$$

Different Approaches

① $20 \quad 20 \quad \frac{2\pi}{3} + \left(\frac{4\pi}{3} \right)$

② $\frac{2}{3}\pi r^2 +$ $\frac{2}{3}\pi r^2 + \frac{1}{2} \times 20\sqrt{3} \times 10$

If answer

• 1251 m² means length of chord was incorrect. Took $10\sqrt{3}$ instead of $20\sqrt{3}$.

• 838 - means only area of major sector was calculated.

3
Marks

b (i) $\left. \begin{aligned} \alpha + \beta &= 3 \\ \alpha \beta &= -13 \end{aligned} \right\} \checkmark$

(ii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2\beta^2} \checkmark$

$$= \frac{9 + 26}{169} = \frac{35}{169} \checkmark$$

1 $\frac{1}{2}$ mark each.

2

Suggested Solutions

Marks

Marker's Comments

(iii) $\alpha + \beta = 3 \therefore \alpha = 3 - \beta$
 $\alpha\beta = -13$

$(3 - \beta)\beta = -13$

$3\beta - \beta^2 = -13$

$\therefore 6\beta - 2\beta^2 = -26$ ✓

1

β is a root

$\therefore \beta^2 - 3\beta - 13 = 0$

$\therefore \beta^2 - 3\beta = 13$

$\therefore 6\beta - 2\beta^2 = 2 \times (-13)$
 $= -26$

c) i) In $\triangle FBE$ and $\triangle GDC$,

$\angle BFE = \angle ECD = 60^\circ$

(alternate angles are equal, $AB \parallel CD$ opp. sides of square) ✓

$\angle FEB = \angle DEC$

(vertically opposite angles are equal) ✓

$\therefore \triangle DEC \parallel \triangle BEF$ (Equiangular) ✓

3

ii) $\angle FCB = 30^\circ$ ($\angle DCB$ is 90°)

$FB = BC \tan 30^\circ$ ✓

$= 1 \left(\frac{1}{\sqrt{3}} \right)$

$= \frac{1}{\sqrt{3}}$

iii) $DC = 1$ and $FB = \frac{1}{\sqrt{3}}$

$\therefore \frac{DC}{FB} = \frac{1}{\frac{1}{\sqrt{3}}} = \sqrt{3}$

$\therefore$ ratio of areas $= (\sqrt{3})^2 : 1^2$
 $= 3 : 1$ ✓

Different approaches

$\frac{\text{Area DEC}}{\text{Area BEF}}$

using

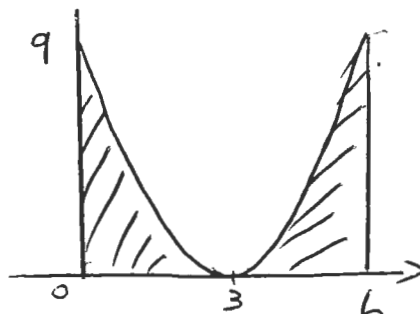
$A = \frac{1}{2} ab \sin C$

Suggested Solutions

Marks

Marker's Comments

d).
Subtract rotation
of shaded area
from cylinder.



$$\text{Volume}_{(\text{Total})} = V_{\text{cylinder}} - V_{(\text{shaded area})}$$

$$= \pi 9^2 \times 6 - \pi \int_0^6 y^2 \cdot dx$$

$$= 486\pi - \pi \int_0^6 ((x-3)^2)^2 dx$$

$$= 486\pi - \pi \int_0^6 (x-3)^4 dx$$

$$= 486\pi - \pi \left[\frac{(x-3)^5}{5} \right]_0^6$$

$$= 486\pi - \frac{2\pi 3^5}{5}$$

$$= 486\pi - \frac{486\pi}{5}$$

$$= \frac{1944\pi}{5}$$

$$\therefore \text{Volume is } \frac{1944\pi}{5} \text{ m}^3$$

$$\text{accepted } 388.8\pi \text{ m}^3$$

$$\text{or } 1221.45 \text{ m}^3$$

✓ V_{cylinder}

✓ $V(x-3)^4$

✓ $\pi \int_0^6$ with

for correct working.

If

• 486π - did not square $(x-3)^2$.

Some did not find volume of the cylinder

• Some expanded $(x-3)^4$ as

$$x^4 - 12x^3 + 54x^2 - 108x + 81$$

MATHEMATICS: Question... 15

[illegible]

MATHEMATICS: Question.....

Suggested Solutions	Marks	Marker's Comments
$500\,000 = 1154\,731.026 - M \frac{(1.0075^{96} - 1)}{0.0075}$ $M = \frac{654\,731.026}{139.856}$ $= 4681.459$ ∴ M is \$4681.46 to the nearest cent	1/2 1/2 1/2	* If they summed 95 terms instead of 96, they got $\frac{654\,731.026}{137.8224951}$ $= 4750.538187$ → 3 1/2 mks
(11) (2) $P = 500\,000$ $r = 12\%$ p.a. $= 1\%$ p.month $M = \$6500$ $A_n = 500\,000(1.01)^n - M(1 + 1.01 + 1.01^2 + \dots + 1.01^{n-1})$ $= 500\,000(1.01)^n - 6500(1 + 1.01 + \dots + 1.01^{n-1})$ $= 500\,000(1.01)^n - 6500 \left[\frac{(1.01)^n - 1}{1.01 - 1} \right]$ $A_n = 500\,000(1.01)^n - 650\,000[1.01^n - 1]$ but $A_n = 0$ $0 = 500\,000(1.01)^n - 650\,000(1.01^n - 1)$ $0 = 500\,000(1.01)^n - 650\,000(1.01^n) + 650\,000$ $150\,000(1.01)^n = 650\,000$ $1.01^n = \frac{13}{3}$ $\log_{1.01} \left(\frac{13}{3} \right) = n$ $n = \frac{\ln(13/3)}{\ln(1.01)}$ $n = 147.365$ ∴ 148 full payments are made.	1/2 1/2 1/2 1/2 1/2 1/2 1/2	* If they used "n-1" in sum of G.P. they got 152 full payments.

MATHEMATICS: Question.....

Suggested Solutions	Marks	Marker's Comments
(P) $A_{148} = 500\,000(1.01)^{148} - 6500(1.01)^{148} - 1$ $= 500\,000(1.01)^{148} - 650\,000(1.01)^{148} + 650\,000$ $= 2180\,385.696 - 2184\,501.404$ $= -4115.708$ $\therefore$ Refund is \$4115.71 * If you did A_{147}, you get 2360.68449 $= 6500 - 2360.68449$ refund = 4139.31551 $= 4139.32$ * refund of \$4123.77 $\Rightarrow$ omk	1/2 1/2 1/2 1 mk, as it doesn't include the interest on the last month!!	

Suggested Solutions

Marks

Marker's Comments

(a) $y = \frac{10}{3+2\sin x} \quad 0 \leq x \leq 2\pi$

$= 10(3+2\sin x)^{-1}$

$\frac{dy}{dx} = -10(3+2\sin x)^{-2} \times 2\cos x$
 $= \frac{-20\cos x}{(3+2\sin x)^2}$

For possible turning points $\frac{dy}{dx} = 0$

i.e. $-20\cos x = 0$

$\cos x = 0$

$\therefore x = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$

Testing the nature of the points
 $(\frac{\pi}{2}, 2)$ and $(\frac{3\pi}{2}, 10)$

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
dy/dx	-20/9	0	20/9	0	-20/9
	\	-	/	-	\

The function is continuous and differentiable throughout.

Since the gradient changes sign at $\frac{\pi}{2}$; $(\frac{\pi}{2}, 2)$ is a relative minimum

Since the gradient changes sign at $\frac{3\pi}{2}$; $(\frac{3\pi}{2}, 10)$ is a relative maximum

When $x=0$, $y = \frac{10}{3+2\sin 0} = \frac{10}{3}$

When $x=2\pi$, $y = \frac{10}{3+2\sin 2\pi} = \frac{10}{3}$

Thus, $(\frac{\pi}{2}, 2)$ is an absolute minimum & $(\frac{3\pi}{2}, 10)$ is an absolute maximum.

Students used the quotient rule and many made errors with $\frac{d(10)}{dx} = 0$

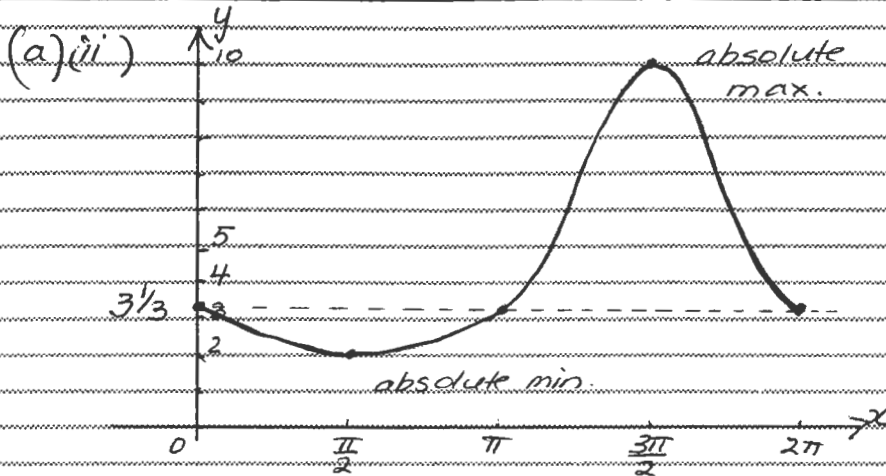
Some students used the second derivative which was more tedious.

3

Suggested Solutions

Marks

Marker's Comments



2

$\frac{1}{2}$ each for endpoints
& $\frac{1}{2}$ each for
max & min.

3

b) (i) $P_A = 500 + 50t$

2

(Errors in making
 $P_A = 500 + (t-1)50$
 $= 450 + 50t$)

3

(ii) (a) Show $P_B = 500e^{-0.3t}$

satisfies $\frac{dP_B}{dt} = -0.3P_B$

$P_B = 500e^{-0.3t}$

$\frac{dP_B}{dt} = 500e^{-0.3t} \times (-0.3)$

$= -0.3 \cdot P_B$ since $P_B = 500e^{-0.3t}$

Thus, $P_B = 500e^{-0.3t}$ satisfies the equation.

$\frac{1}{2}$

$\frac{1}{2}$

(students failed
to justify the
final step)

(b) When $t=6$, find P_B

$P_B = 500e^{-0.3 \times 6}$
 $= 500e^{-1.8}$
 $= 82.649...$

$\frac{1}{2}$

$\therefore P_B = 83$ (to the nearest integer)

$\frac{1}{2}$

82?

(c) $P_C = 100(5 + t - \frac{t^2}{4})$
 $= 500 + 100t - 25t^2$

$\frac{dP_C}{dt} = 100 - 50t$

We can determine a maximum when

$\frac{dP_C}{dt} = 0$ and $\frac{d^2P_C}{dt^2} < 0$

i.e. $100 - 50t = 0$
 $t = 2$

$\frac{1}{2}$

$\frac{1}{2}$

Alternatively,

$t = \frac{-(-4)}{2 \times 1} = 2$

(1mk)

using $t = \frac{-b}{2a}$

because the
max. lies on the
axis of the
parabola.

Suggested Solutions

Marks

Marker's Comments

$$\frac{d^2 P_c}{dt^2} = -50$$

i.e. $\frac{d^2 P_c}{dt^2} < 0$ and concave down

when $t = 2$

Thus, there is a maximum after 2 years

$$P_c(\text{max}) = 100 \left(5 + 2 - \frac{2^2}{4} \right)$$

$$= 600$$

Maximum population of P_c is 600 people.

(8.) A town is deemed unviable if the population goes below 50. as $P_A = 500 + 50t$ it continues to increase for $t \leq 10$, so does not drop below 50.

$$P_B = 500e^{-0.3t} = 50$$

$$e^{-0.3t} = \frac{1}{10}$$

$$e^{0.3t} = 10$$

$$0.3t = \ln 10$$

$$t = \frac{10}{3} \ln 10$$

$$= 7.67...$$

$$P_c = 500 + 100t - 25t^2 = 50$$

$$t^2 - 4t - 18 = 0$$

$$t = \frac{-(-4) \pm \sqrt{(-4)^2 - 4 \cdot 1 \cdot (-18)}}{2 \times 1}$$

$$= \frac{4 \pm \sqrt{88}}{2}$$

$$t = \frac{4 + 2\sqrt{22}}{2}, \text{ as } t > 0$$

$$\therefore t = 2 + \sqrt{22}$$

$$= 6.6904...$$

The population of C will drop below 50 sooner than town B, thus town C will be the first to close

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

Alternatively,

$$a = -25$$

$$a < 0$$

thus, curve is concave down

and the vertex gives a maximum.

[4]

[3]